

# Lumana Shakya

Ann Arbor, MI

Email: [lshakya@umich.edu](mailto:lshakya@umich.edu)  
Website: [lumanashakya.github.io](https://lumanashakya.github.io)

## ACADEMIC POSITIONS:

**Postdoctoral Research Fellow**, School for Environment and Sustainability, University of Michigan, Ann Arbor, MI Aug 2026–Present

## EDUCATION

**Ph.D. in Economics**, North Carolina State University, Raleigh, NC Aug 2021–July 2026

- Dissertation: “*Essays on Climate Change and Health*”
- Committee: Raymond Guiteras, Robyn Meeks, Zachary Brown, Zheng Li & Zhenxuan Wang

**M.A. in Economics**, North Carolina State University, Raleigh, NC Aug 2021–May 2023

**B.A. in Economics, minor in Mathematics**, Ramapo College of New Jersey, Mahwah, NJ Sept 2017–May 2021  
(*Summa Cum Laude*)

## RESEARCH FIELDS

Environmental Economics, Development Economics, Global Health

## PUBLICATIONS

The effects of orphanhood and lack of parental care on child vaccination: analyses of 189 cross-sectional household surveys from 82 low- and middle-income countries, 2005 to 2022. **eClinicalMedicine**, 85, 103314, July 2025, doi: 10.1016/j.eclinm.2025.103314. (with A. Bolgrien, B. Damte, M. Johri, R. A. Medeiros, and M. Munir)

## WORKING PAPERS

Climate Change and Energy Consumption in Bangladesh (**Job Market Paper**)

Temperature Extremes, Electricity Access, and All-Cause Mortality in Bangladesh

Routine Disasters: Floods, Human Capital and Adaptation in Bangladesh (with R. Guiteras, A. Jina, and A.M. Mobarak)

## WORKS IN PROGRESS

Exploring orphanhood and health and education outcomes for children using MICS

## CONFERENCES AND SEMINARS

|                                                                                     |      |
|-------------------------------------------------------------------------------------|------|
| Southern Economics Association (SEA) Annual Meeting, Houston, TX                    | 2026 |
| American Economics Association (AEA) Annual Meeting, Philadelphia, PA               | 2026 |
| Camp Resources XXXI, Asheville, NC                                                  | 2025 |
| NCSU Graduate Student Research Symposium, Raleigh, NC                               | 2025 |
| Agricultural & Applied Economics Association (AAEA) Annual Meeting, New Orleans, LA | 2024 |

## FELLOWSHIPS, AWARDS AND GRANTS

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| NC State's Center for Environmental Resource Economic Policy (CEnREP) Summer Research Grant     | 2025 |
| AAEA Teaching Academy Travel Grant                                                              | 2024 |
| NC State University Provost's Doctoral Fellowship                                               | 2021 |
| Award for Exemplary Volunteer Work at NJ SBDC, Consulate General of Nepal, NY                   | 2021 |
| American Federation of Teachers (AFT) Graduate School Scholarship, Ramapo College of New Jersey | 2021 |
| NJ Collegiate Business Administration Society Honor Society                                     | 2021 |

## TEACHING EXPERIENCE

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| <b>Independent Instructor</b> , North Carolina State University                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jan 2024 – Present |
| <ul style="list-style-type: none"><li>• ARE 201 Introduction to Agricultural and Resource Economics: Spring 24, Fall 24, Spring 26 (Class sizes: 30-200 students)</li></ul>                                                                                                                                                                                                                                                                                                                        |                    |
| <b>Graduate Teaching Assistant</b> , North Carolina State University                                                                                                                                                                                                                                                                                                                                                                                                                               | Aug 2021 – Present |
| <ul style="list-style-type: none"><li>• EC 205 Fundamentals of Economics (Micro and Macro): Fall 22</li><li>• EC 201 Principles of Microeconomics, Fall 23</li><li>• ARE 336 Introduction to Resource and Environmental Economics: Fall 25</li><li>• ARE 312 Agribusiness Marketing: Spring 25, Spring 26</li><li>• ARE 104 Agricultural Business Management: Spring 26</li><li>• ARE 412 Advanced Agribusiness Marketing: Spring 26</li><li>• ARE 455 Agribusiness Analytics: Spring 24</li></ul> |                    |

## PREVIOUS PROFESSIONAL EMPLOYMENT

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| <b>Summer Fellow</b> , Institute for Social Research & Data Innovation, University of Minnesota                                                                                                                                                                                                                                                                                                                                                   | June 2024 – Aug 2024 |
| <ul style="list-style-type: none"><li>• Project: <i>Childhood immunization in IPUMS Global Health</i> (Advisor: Dr. Anna Bolgrien)</li><li>• Documented patterns in global childhood vaccination schedules across more than 100 countries to facilitate the release of vaccination data on IPUMS-MICS.</li><li>• Created new IPUMS-MICS variables to harmonize data management across time and place for global public health research.</li></ul> |                      |
| <b>Summer Research Assistant</b> , Duke University                                                                                                                                                                                                                                                                                                                                                                                                | June 2023 – Aug 2023 |
| <ul style="list-style-type: none"><li>• Project: <i>Randomized Control Trial (RCT) examining uptake of electric induction stove in Nepal</i> (Advisor: Dr. Robyn Meeks)</li><li>• Assisted in the design of the randomized experiment and in the synthesis of literature on clean cooking and adaptive technologies in Nepal and other low- and middle-income countries.</li></ul>                                                                |                      |
| <b>Business Consultant</b> (Volunteer), NJ Small Business Development Center (SBDC)                                                                                                                                                                                                                                                                                                                                                               | Jan 2021 – May 2021  |
| <ul style="list-style-type: none"><li>• Assisted over a dozen small businesses coping with COVID-19 impacts.</li><li>• Conducted market research to assist in drafting business plans and building financial forecasts.</li></ul>                                                                                                                                                                                                                 |                      |
| <b>Financial Planning &amp; Analysis</b> (Part-time), Promotion in Motion/ P.I.M. Brands                                                                                                                                                                                                                                                                                                                                                          | May 2019 – Dec 2020  |
| <ul style="list-style-type: none"><li>• Assisted in preparing monthly financial report packages and yearly budgets/forecasts for senior management.</li><li>• Provided analytical support to the Sales and Marketing departments and built financial models for a new subsidiary.</li></ul>                                                                                                                                                       |                      |

## SERVICE

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Referee: Journal of Environmental Economics and Management (JEEM)

## OTHER INFORMATION

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Technical Skills: Stata, R, Python, Google Earth Engine, EViews,  $\text{\LaTeX}$ , Matlab

Languages: English (Proficient), Nepalese (Native), Newari (Native), Hindi (Fluent)

## REFERENCES

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Raymond Guiteras (Advisor)  
Associate Professor  
Dept. of Agricultural & Resource Economics  
NC State University, Raleigh, NC  
[rpguiter@ncsu.edu](mailto:rpguiter@ncsu.edu)

Zachary Brown  
Associate Professor  
Dept. of Agricultural & Resource Economics  
NC State University, Raleigh, NC  
[zsbrown2@ncsu.edu](mailto:zsbrown2@ncsu.edu)

Zhenxuan Wang  
Assistant Professor  
Dept. of Agricultural & Resource Economics  
NC State University, Raleigh, NC  
[zhenxuan.wang@ncsu.edu](mailto:zhenxuan.wang@ncsu.edu)